

Kalkulator Huruf

By: Febrian Ananda T

Time Limit	1 s
Memory Limit	256 MB



Alvino adalah seorang filsuf asal Indonesia yang sedang merenungkan sistem bilangan. Ia bertanya-tanya, "Mengapa sistem bilangan kita berbeda dengan sistem huruf kita? Padahal, keduanya sama-sama krusial bagi kehidupan kita." Dari pemikiran ini, muncul sebuah gagasan: "Bagaimana jika sistem huruf digunakan persis seperti sistem bilangan?"

Gagasannya adalah agar huruf $a - z$ mewakili angka $1 - 26$ dan huruf $A - Z$ mewakili angka $27 - 52$. Alvino membayangkan sebuah sistem di mana operasi penjumlahan, pengurangan, perkalian, pembagian, dan modulo dapat dilakukan sama seperti pada angka. Jika hasilnya melebihi 52, urutan huruf dimulai kembali dari awal; jika hasilnya kurang dari 1, urutan dimulai dari akhir. Sebagai contoh, angka 53 diwakili oleh huruf 'a', dan angka 0 diwakili oleh huruf 'Z'.

Namun, Alvino masih merasa ragu dengan sistem tersebut; ia ingin menguji prototipenya terlebih dahulu untuk melihat apakah gagasannya benar-benar brilian. Oleh karena itu, ia meminta kamu, sahabatnya, untuk membuat prototipe tersebut

Format Masukan:

Baris pertama berisi: $a \text{ op } b$. Dimana a dan b adalah huruf $a - z$ atau $A - Z$ dan op adalah operator matematika $+$, $-$, $*$, $/$, $\%$.

+	-	*	/	%
Tambah	Kurang	Perkalian	Pembagian integer (bulatkan ke bawah)	Modulo

Format Keluaran:

Hasil dalam bentuk huruf $a - z$ atau $A - Z$

Sample Input 1

e + b

Sample Output 1

g

Penjelasan: huruf e dan b masing-masing memiliki nilai ASCII 101, 98.

```
printf("%d %d\n", 'e', 'b');  
printf("%c %c\n", 101, 98);  
printf("%c", (5 + 2)+'a'-1);
```

```
101 98  
e b  
g
```

Sementara di sistem baru Alvino huruf tersebut bernilai 5 dan 2 sehingga hasil operasi e + b adalah 7 alias huruf g.

Sample Input 2

f * f

Sample Output 2

J

Penjelasan: huruf f di sistem baru Alvino bernilai 6. 6×6 adalah 36 alias huruf J

Sample Input 3

a - a

Sample Output 3

Z

Sample Input 4

a - b

Sample Output 4

Y

ASCII TABLE

Decimal	Hexadecimal	Binary	Octal	Char	Decimal	Hexadecimal	Binary	Octal	Char	Decimal	Hexadecimal	Binary	Octal	Char
0	0	0	0	[NULL]	48	30	110000	60	0	96	60	1100000	140	`
1	1	1	1	[START OF HEADING]	49	31	110001	61	1	97	61	1100001	141	a
2	2	10	2	[START OF TEXT]	50	32	110010	62	2	98	62	1100010	142	b
3	3	11	3	[END OF TEXT]	51	33	110011	63	3	99	63	1100011	143	c
4	4	100	4	[END OF TRANSMISSION]	52	34	110100	64	4	100	64	1100100	144	d
5	5	101	5	[ENQUIRY]	53	35	110101	65	5	101	65	1100101	145	e
6	6	110	6	[ACKNOWLEDGE]	54	36	110110	66	6	102	66	1100110	146	f
7	7	111	7	[BELL]	55	37	110111	67	7	103	67	1100111	147	g
8	8	1000	10	[BACKSPACE]	56	38	111000	70	8	104	68	1101000	150	h
9	9	1001	11	[HORIZONTAL TAB]	57	39	111001	71	9	105	69	1101001	151	i
10	A	1010	12	[LINE FEED]	58	3A	111010	72	:	106	6A	1101010	152	j
11	B	1011	13	[VERTICAL TAB]	59	3B	111011	73	;	107	6B	1101011	153	k
12	C	1100	14	[FORM FEED]	60	3C	111100	74	<	108	6C	1101100	154	l
13	D	1101	15	[CARRIAGE RETURN]	61	3D	111101	75	=	109	6D	1101101	155	m
14	E	1110	16	[SHIFT OUT]	62	3E	111110	76	>	110	6E	1101110	156	n
15	F	1111	17	[SHIFT IN]	63	3F	111111	77	?	111	6F	1101111	157	o
16	10	10000	20	[DATA LINK ESCAPE]	64	40	1000000	100	@	112	70	1110000	160	p
17	11	10001	21	[DEVICE CONTROL 1]	65	41	1000001	101	A	113	71	1110001	161	q
18	12	10010	22	[DEVICE CONTROL 2]	66	42	1000010	102	B	114	72	1110010	162	r
19	13	10011	23	[DEVICE CONTROL 3]	67	43	1000011	103	C	115	73	1110011	163	s
20	14	10100	24	[DEVICE CONTROL 4]	68	44	1000100	104	D	116	74	1110100	164	t
21	15	10101	25	[NEGATIVE ACKNOWLEDGE]	69	45	1000101	105	E	117	75	1110101	165	u
22	16	10110	26	[SYNCHRONOUS IDLE]	70	46	1000110	106	F	118	76	1110110	166	v
23	17	10111	27	[END OF TRANS. BLOCK]	71	47	1000111	107	G	119	77	1110111	167	w
24	18	11000	30	[CANCEL]	72	48	1001000	110	H	120	78	1111000	170	x
25	19	11001	31	[END OF MEDIUM]	73	49	1001001	111	I	121	79	1111001	171	y
26	1A	11010	32	[SUBSTITUTE]	74	4A	1001010	112	J	122	7A	1111010	172	z
27	1B	11011	33	[ESCAPE]	75	4B	1001011	113	K	123	7B	1111011	173	{
28	1C	11100	34	[FILE SEPARATOR]	76	4C	1001100	114	L	124	7C	1111100	174	
29	1D	11101	35	[GROUP SEPARATOR]	77	4D	1001101	115	M	125	7D	1111101	175	}
30	1E	11110	36	[RECORD SEPARATOR]	78	4E	1001110	116	N	126	7E	1111110	176	~
31	1F	11111	37	[UNIT SEPARATOR]	79	4F	1001111	117	O	127	7F	1111111	177	[DEL]
32	20	100000	40	[SPACE]	80	50	1010000	120	P					
33	21	100001	41	!	81	51	1010001	121	Q					
34	22	100010	42	"	82	52	1010010	122	R					
35	23	100011	43	#	83	53	1010011	123	S					
36	24	100100	44	\$	84	54	1010100	124	T					
37	25	100101	45	%	85	55	1010101	125	U					
38	26	100110	46	&	86	56	1010110	126	V					
39	27	100111	47	'	87	57	1010111	127	W					
40	28	101000	50	(	88	58	1011000	130	X					
41	29	101001	51	)	89	59	1011001	131	Y					
42	2A	101010	52	*	90	5A	1011010	132	Z					
43	2B	101011	53	+	91	5B	1011011	133	[					
44	2C	101100	54	,	92	5C	1011100	134	\					
45	2D	101101	55	-	93	5D	1011101	135	]					
46	2E	101110	56	.	94	5E	1011110	136	^					
47	2F	101111	57	/	95	5F	1011111	137	_					

DILARANG MENGGUNAKAN ARRAY, LOOP ATAU MATERI LAIN DI LUAR MODUL 0, 1, DAN X.

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Alvino, an Indonesian philosopher, was reflecting on number systems. He wondered, “Why is our number system different from our writing system? After all, both are equally crucial to our lives.” From this thought, an idea emerged: “What if the writing system were used just like the number system?”

The idea is for the letters $a - z$ to represent the numbers $1 - 26$ and the letters $A - Z$ to represent the numbers $27 - 52$. Alvino envisions a system that functions like a standard number system, capable of performing addition, subtraction, multiplication, division, and modulo operations. If a result exceeds 52, the lettering sequence restarts from the beginning; if the result is less than 1, the sequence starts from the end. For example, the number 53 is represented by the letter ‘a’, and the number 0 is represented by the letter ‘Z’.

However, Alvino remains skeptical about this system; he wants to test a prototype first to see whether his idea is brilliant or not. Therefore, Alvino is asking for your help, as his good friend, to build that prototype.

Input Format:

The first line contains: $a \text{ op } b$. Where a , and b are the letters $a - z$ or $A - Z$ and op are the mathematical operators $+$, $-$, $*$, $/$, $\%$.

+	-	*	/	%
Addition	Subtraction	Multiplication	Integer division (Round down)	Modulo

Output Format:

The results are in the form of letters $a - z$ or $A - Z$

Input Sample 1

e + b

Output Sample 1

g

Explanation: The letters 'e' and 'b' have ASCII values of 101 and 98, respectively.

```
printf("%d %d\n", 'e', 'b');  
printf("%c %c\n", 101, 98);  
printf("%c", (5 + 2)+'a'-1);
```

```
101 98  
e b  
g
```

Meanwhile, in Alvino's new system, those letters have values of 5 and 2, so the result of the operation $e + b$ is 7—or the letter g.

Input Sample 2

f * f

Output Sample 2

J

Explanation: In Alvino's new system, the letter 'f' has a value of 6. 6×6 is 36, which corresponds to the letter 'J'.

Input Sample 3

a - a

Output Sample 3

Z

Input Sample 4

a - b

Output Sample 4

Y

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36	24	100100	44	\$	84	54	1010100	124	T					
37	25	100101	45	%	85	55	1010101	125	U					
38	26	100110	46	&	86	56	1010110	126	V					
39	27	100111	47	'	87	57	1010111	127	W					
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41	29	101001	51	)	89	59	1011001	131	Y					
42	2A	101010	52	*	90	5A	1011010	132	Z					
43	2B	101011	53	+	91	5B	1011011	133	[					
44	2C	101100	54	,	92	5C	1011100	134	\					
45	2D	101101	55	-	93	5D	1011101	135	]					
46	2E	101110	56	.	94	5E	1011110	136	^					
47	2F	101111	57	/	95	5F	1011111	137	_					

DO NOT USE ANY LOOPS OR ARRAYS OR ANY OTHER MATERIALS OUTSIDE OF MODULE 0, 1, AND X